U-Net Baseline Implementation for Dynamic Human Movement Segmentation

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ABSTRACT

Automated segmentation of human figures in dynamic motion is a crucial prerequisite for advanced biomechanical analysis and sports science. This study details the development of a convolutional neural network baseline, utilizing the U-Net architecture, designed specifically to isolate human subjects from complex backgrounds. To overcome the spatial bias typical of standard portrait datasets, the model was trained on the highly dynamic TikTok Dances dataset. Optimization was achieved through a `combined_loss` function, integrating Binary Cross-Entropy and Dice Loss to enforce strict edge adherence. By implementing a `ReduceLROnPlateau` scheduler over 30 epochs, the model achieved efficient convergence with a Validation Dice Score of 0.9506 and a Validation IoU of 0.9061. This demonstrates the model's proficiency in capturing precise anatomical silhouettes during extreme movement articulations.

1. Introduction

Extracting precise human silhouettes from unconstrained backgrounds remains a foundational challenge in computer vision. Traditional segmentation algorithms often struggle to maintain edge fidelity when subjects engage in rapid, complex movements. Recent advancements have explored the integration of deep learning in sports analytics, specifically for monitoring athlete motion quality [1] and human activity recognition within sensor-based environments [2]. However, standard models frequently overfit to static poses, resulting in fragmented predictions during extreme physical maneuvers. This project addresses these limitations by developing a robust baseline using the U-Net architecture. By training the network on a dataset featuring dynamic dance movements, the model is forced to learn scale-invariant and pose-agnostic anatomical representations.

2. Methodology and Network Architecture

The architecture follows the fundamental U-Net topology [3], characterized by a symmetric encoder-decoder structure. The contracting path (encoder) systematically reduces the spatial dimensions of the input image through four stages of increasing channel depth (64, 128, 256, and 512 channels) to capture global semantic context. Conversely, the expanding path (decoder) upsamples these feature maps to restore the original spatial resolution required for pixel-wise classification. The critical innovation of this architecture lies in its skip connections, which directly transfer high-resolution spatial details from the encoder to the decoder. This mechanism ensures that fine morphological details, such as distal limbs and joints, are accurately reconstructed in the final binary mask. Additionally, stochastic horizontal flipping was applied to enhance the model's robustness against subject orientation. While recent hybrid models have begun integrating advanced attention modules [4], this study prioritizes a pure conventional baseline for computational efficiency in medical and general image segmentation [5].

3. Technical Implementation

The technical implementation of the proposed U-Net baseline is designed to provide a robust pipeline for precise pixel-wise classification of human figures. Developed using the PyTorch framework, the methodology prioritizes a modular approach to ensure both scalability and computational efficiency.

3.1 Architecture Definition

The network's symmetry is constructed using modular convolutional blocks. The progression from downsampling to upsampling utilizes concatenated feature maps to retain spatial boundaries and prevent information loss during the compression phase. Each block consists of dual 3x3 convolutions followed by Batch Normalization and ReLU activation to ensure stable gradient flow. The architecture reaches a bottleneck depth of 512 channels, providing a rich semantic representation before the expanding path restores the spatial resolution to the original 128x128 dimensions.

```
class UNet(nn.Module):
    def __init__(self, in_channels=3, out_channels=1):
        super().__init__()
        self.down1 = DoubleConv(in_channels, 64)
        self.down2 = DoubleConv(64, 128)
        self.down3 = DoubleConv(128, 256)
        self.down4 = DoubleConv(256, 512)

        self.up1 = nn.ConvTranspose2d(512, 256, kernel_size=2, stride=2)
        self.up_conv1 = DoubleConv(512, 256)

        self.up2 = nn.ConvTranspose2d(256, 128, kernel_size=2, stride=2)
        self.up_conv2 = DoubleConv(256, 128)

        self.up3 = nn.ConvTranspose2d(128, 64, kernel_size=2, stride=2)
        self.up_conv3 = DoubleConv(128, 64)

        self.out_conv = nn.Conv2d(64, out_channels, kernel_size=1)
```

3.2 Hybrid Loss Formulation

Optimizing the network solely on Binary Cross-Entropy (BCE) frequently generates soft boundary artifacts, especially in complex movement poses. To enforce strict edge delineation, the network was optimized using a hybrid loss function. The BCE component evaluates pixel-wise probability:

$$L_{BCE} = -\frac{1}{N}\sum_{i=1}^{N}[y_i \log(p_i) + (1-y_i)\log(1-p_i)]$$

To complement this, the Dice Coefficient is integrated to penalize structural mismatches and maximize the spatial overlap between the prediction and the ground truth:

$$L_{Dice} = 1 - \frac{2\sum_{i=1}^{N} p_i y_i}{\sum_{i=1}^{N} p_i + \sum_{i=1}^{N} y_i}$$

The final objective function minimized during training is the linear combination of both components ($L_{BCE} + L_{Dice}$), ensuring both localized pixel accuracy and global anatomical consistency.

```
def combined_loss(pred, target):
    bce = nn.BCEWithLogitsLoss()(pred, target)
    pred_sig = torch.sigmoid(pred)
    smooth = 1e-6
    inter = (pred_sig * target).sum()
    dice = 1 - (2. * inter + smooth) / (pred_sig.sum() + target.sum() + smooth)
    return bce + dice
```

3.3 Optimization Strategy

Standard optimization is often insufficient for capturing fine morphological details in high-variance datasets. Therefore, the training process employed the Adam optimizer with an initial learning rate of 1×10^{-4}. To ensure efficient convergence and prevent the model from getting stuck in local minima, a `ReduceLROnPlateau` scheduler was implemented with a patience of 3 epochs. This allows the system to automatically adjust the learning pace whenever the validation loss plateaus, leading to a more refined and stable final model.

3.4 Training Protocol and Dataset

To directly address the necessity of modeling dynamic movement, the TikTok Dances Segmentation Dataset (2,615 samples) was selected over conventional portrait datasets. This dataset provides the high variance in body articulation required for robust inference. The training was conducted for 30 epochs with an 80-20 split for training and validation. Additionally, stochastic horizontal flipping was applied as a data augmentation technique to enhance the model's invariance to the subject's lateral orientation.

4. Results and Analysis

Quantitative evaluation indicates that the model achieved rapid and stable convergence over the 30-epoch training schedule. The training loss decreased significantly from an initial 1.0354 to a final value of 0.0581, as

illustrated in Figure 1. Upon completion, the model yielded a Validation Dice Score of 0.9506 and a Validation IoU of 0.9061.

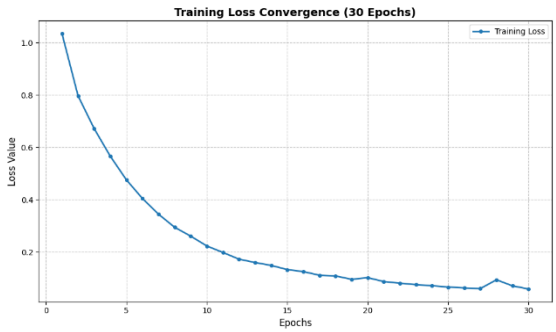

Figure 1 Training loss convergence curve showing stability over 30 epochs.

Qualitative analysis confirms the model's proficiency in handling dynamic poses. As shown in Figure 2, the U-Net successfully isolated subjects without fragmentation.

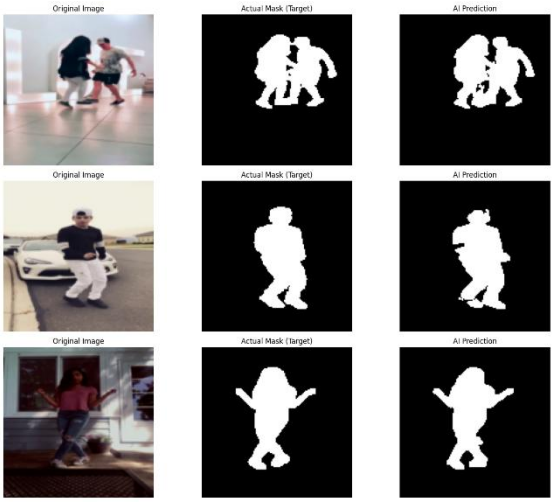

Figure 2 Qualitative segmentation results showing the model's prediction after training

To validate practical utility, the model was deployed as an interactive application using `Streamlit`. Figure 3 displays the web interface, while Figure 4 demonstrates the model's ability to perform real-time inference on new user-uploaded images.

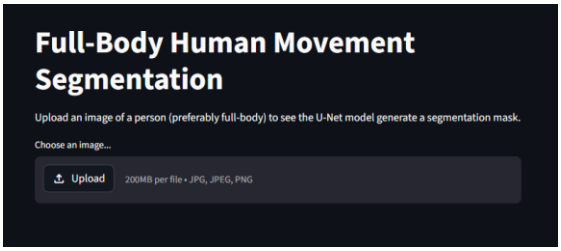

Figure 3 Interactive Streamlit interface for human segmentation.

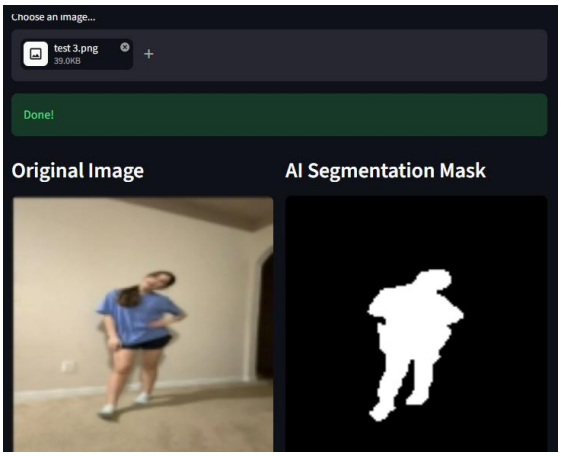

Figure 4 Real-time testing result demonstrated within the Streamlit application.

4.1 Challenges and Key Learnings

Throughout the development, several technical challenges provided valuable learning experiences:

- Edge Delineation & Loss Selection: Initial experiments with pure BCE resulted in "ghosting" artifacts around limbs. Implementing the hybrid combined_loss taught me that complex morphological segmentation requires a balance between pixel-level probability and structural overlap.
- Dataset Homogeneity & Scale Bias: While the TikTok dataset provides excellent pose variety, its composition is highly homogeneous (mostly full-body shots with centered subjects). Testing with out-of-distribution images such as half-body portraits, subjects wearing loose-fitting graduation

gowns, or low-contrast backgrounds revealed a generalization limit.

- Inference Realities: I observed that the model struggles when clothing colors blend with the background or when the subject holds external objects. This highlights the need for more diverse data augmentation (color jittering, random cropping) and potentially the integration of attention mechanisms, similar to "Stunet-MHA," to better focus on relevant semantic features.

5. Conclusion

The implemented U-Net baseline successfully provides high-fidelity segmentation for dynamic human movement. The strategic combination of a motion-centric dataset, `combined_loss` optimization, and adaptive learning rate scheduling established a highly reliable model. With a Dice Score of 0.9506, this research provides a solid foundation for future work in automated athlete motion quality detection and advanced sports analytics.

6. References

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